



Percent Change in Two Stages

Grades 9-12 • SAT / ACT mathematics foundations

Problem-Solving and Data Analysis • Distinguish additive and multiplicative percent changes.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Distinguish additive and multiplicative percent changes.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. A price of \$120 rises 20% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

Multiply successive factors: $120 \times 1.2 \times 0.75 = 108.00$. Net change = $(108/120 - 1) \times 100\% = -10\%$.

Model 2. A price of \$160 rises 20% and then falls 20% from the increased price. Find the final price and the net percent change from the original.

Multiply successive factors: $160 \times 1.2 \times 0.8 = 153.60$. Net change = $(153.6/160 - 1) \times 100\% = -4\%$.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A price of \$160 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

2. A price of \$80 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

3. A price of \$120 rises 10% and then falls 10% from the increased price. Find the final price and the net percent change from the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A price of \$160 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

5. A price of \$240 rises 25% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

6. A price of \$240 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A price of \$120 rises 25% and then falls 10% from the increased price. Find the final price and the net percent change from the original.

8. A price of \$120 rises 25% and then falls 20% from the increased price. Find the final price and the net percent change from the original.

9. A price of \$80 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A price of \$120 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

11. A price of \$160 rises 10% and then falls 10% from the increased price. Find the final price and the net percent change from the original.

12. A price of \$80 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A price of \$80 rises 25% and then falls 20% from the increased price. Find the final price and the net percent change from the original.

14. A price of \$240 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

15. A price of \$120 rises 20% and then falls 10% from the increased price. Find the final price and the net percent change from the original.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A price of \$80 rises 20% and then falls 20% from the increased price. Find the final price and the net percent change from the original.



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A price of \$80 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

2. A price of \$160 rises 20% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A price of \$120 rises 10% and then falls 20% from the increased price. Find the final price and the net percent change from the original.

4. A price of \$120 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A price of \$120 rises 20% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

6. A price of \$160 rises 20% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A price of \$160 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

2. A price of \$240 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A price of \$240 rises 10% and then falls 25% from the increased price. Find the final price and the net percent change from the original.

4. A price of \$80 rises 10% and then falls 20% from the increased price. Find the final price and the net percent change from the original.



Practice worked solutions

1. \$132.00; -35/2%

Multiply successive factors: $160 \times 1.1 \times 0.75 = 132.00$. Net change = $(132/160 - 1) \times 100\% = -35/2\%$.

2. \$66.00; -35/2%

Multiply successive factors: $80 \times 1.1 \times 0.75 = 66.00$. Net change = $(66/80 - 1) \times 100\% = -35/2\%$.

3. \$118.80; -1%

Multiply successive factors: $120 \times 1.1 \times 0.9 = 118.80$. Net change = $(594/5/120 - 1) \times 100\% = -1\%$.

4. \$132.00; -35/2%

Multiply successive factors: $160 \times 1.1 \times 0.75 = 132.00$. Net change = $(132/160 - 1) \times 100\% = -35/2\%$.

Practice worked solutions

5. \$225.00; -25/4%

Multiply successive factors: $240 \times 1.25 \times 0.75 = 225.00$. Net change = $(225/240 - 1) \times 100\% = -25/4\%$.

6. \$198.00; -35/2%

Multiply successive factors: $240 \times 1.1 \times 0.75 = 198.00$. Net change = $(198/240 - 1) \times 100\% = -35/2\%$.

7. \$135.00; 25/2%

Multiply successive factors: $120 \times 1.25 \times 0.9 = 135.00$. Net change = $(135/120 - 1) \times 100\% = 25/2\%$.

8. \$120.00; 0%

Multiply successive factors: $120 \times 1.25 \times 0.8 = 120.00$. Net change = $(120/120 - 1) \times 100\% = 0\%$.

Practice worked solutions

9. \$66.00; -35/2%

Multiply successive factors: $80 \times 1.1 \times 0.75 = 66.00$. Net change = $(66/80 - 1) \times 100\% = -35/2\%$.

10. \$99.00; -35/2%

Multiply successive factors: $120 \times 1.1 \times 0.75 = 99.00$. Net change = $(99/120 - 1) \times 100\% = -35/2\%$.

11. \$158.40; -1%

Multiply successive factors: $160 \times 1.1 \times 0.9 = 158.40$. Net change = $(792/5/160 - 1) \times 100\% = -1\%$.

12. \$66.00; -35/2%

Multiply successive factors: $80 \times 1.1 \times 0.75 = 66.00$. Net change = $(66/80 - 1) \times 100\% = -35/2\%$.

Practice worked solutions

13. \$80.00; 0%

Multiply successive factors: $80 \times 1.25 \times 0.8 = 80.00$. Net change = $(80/80 - 1) \times 100\% = 0\%$.

14. \$198.00; -35/2%

Multiply successive factors: $240 \times 1.1 \times 0.75 = 198.00$. Net change = $(198/240 - 1) \times 100\% = -35/2\%$.

15. \$129.60; 8%

Multiply successive factors: $120 \times 1.2 \times 0.9 = 129.60$. Net change = $(648/5/120 - 1) \times 100\% = 8\%$.

16. \$76.80; -4%

Multiply successive factors: $80 \times 1.2 \times 0.8 = 76.80$. Net change = $(384/5/80 - 1) \times 100\% = -4\%$.

Transfer worked solutions

1. \$66.00; -35/2%

Multiply successive factors: $80 \times 1.1 \times 0.75 = 66.00$. Net change = $(66/80 - 1) \times 100\% = -35/2\%$.

2. \$144.00; -10%

Multiply successive factors: $160 \times 1.2 \times 0.75 = 144.00$. Net change = $(144/160 - 1) \times 100\% = -10\%$.

3. \$105.60; -12%

Multiply successive factors: $120 \times 1.1 \times 0.8 = 105.60$. Net change = $(528/5/120 - 1) \times 100\% = -12\%$.

4. \$99.00; -35/2%

Multiply successive factors: $120 \times 1.1 \times 0.75 = 99.00$. Net change = $(99/120 - 1) \times 100\% = -35/2\%$.

Transfer worked solutions

5. \$108.00; -10%

Multiply successive factors: $120 \times 1.2 \times 0.75 = 108.00$. Net change = $(108/120 - 1) \times 100\% = -10\%$.

6. \$144.00; -10%

Multiply successive factors: $160 \times 1.2 \times 0.75 = 144.00$. Net change = $(144/160 - 1) \times 100\% = -10\%$.

Exit-ticket worked solutions

1. \$132.00; -35/2%

Multiply successive factors: $160 \times 1.1 \times 0.75 = 132.00$. Net change = $(132/160 - 1) \times 100\% = -35/2\%$.

2. \$198.00; -35/2%

Multiply successive factors: $240 \times 1.1 \times 0.75 = 198.00$. Net change = $(198/240 - 1) \times 100\% = -35/2\%$.

3. \$198.00; -35/2%

Multiply successive factors: $240 \times 1.1 \times 0.75 = 198.00$. Net change = $(198/240 - 1) \times 100\% = -35/2\%$.

4. \$70.40; -12%

Multiply successive factors: $80 \times 1.1 \times 0.8 = 70.40$. Net change = $(352/5/80 - 1) \times 100\% = -12\%$.

Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Distinguish additive and multiplicative percent changes. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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